

HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by background redundancy and extreme scale variation. While spatial selective computation mitigates arithmetic overhead by routing candidate foreground regions, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a dual-evidence framework featuring semantic-spectral routing and coverage-constrained spatial optimization. HESOD couples high-level semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ spatial complexity to rescue faint targets under an adaptive evidence preservation principle. To guarantee recall safety, an object-level soft-coverage loss directly optimizes multi-cell joint coverage probability across all ground-truth instances. Combined with an inverted residual decoupled head and scale-aware Wasserstein regression, HESOD advances mAP@.5 by +0.7 pp on UAVDT and +2.4 pp on SeaPerson while cutting parameters by 27.6% with negligible FLOPs overhead, establishing a superior accuracy-efficiency Pareto frontier for edge platforms.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, environmental monitoring, and search-and-rescue [1, 2, 4]. In these settings, micro targets typically occupy fewer than 16×16 pixels amidst vast background areas ($> 70\text{--}80\%$), making dense uniform convolution computationally prohibitive on edge platforms [3, 5].

Spatial selective computation mitigates arithmetic redundancy by routing processing strictly to candidate foreground regions [3, 6, 7]. However, existing selectors exhibit two critical vulnerabilities: (i) *Evidence fragility*: relying solely on deep semantic activations leads to severe feature collapse for micro targets after stem downsampling, causing irrecoverable false-negative pruning under asymmetric selection cost [3, 8];

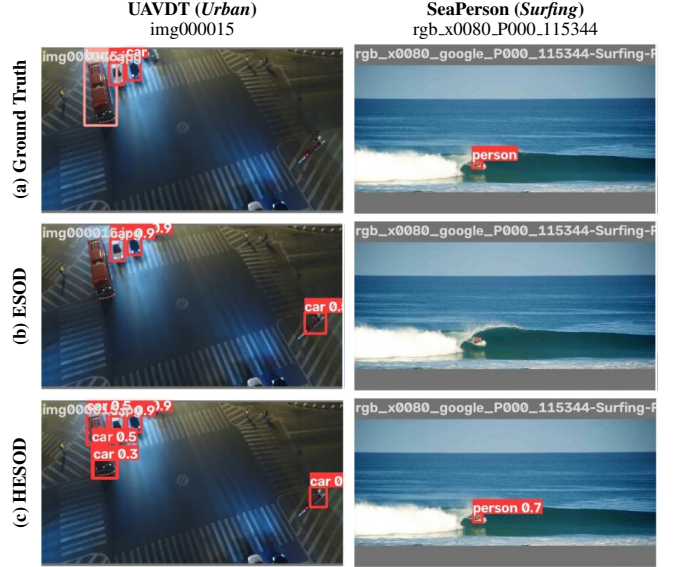


Fig. 1. Qualitative comparison on UAVDT and SeaPerson.

and (ii) *Coverage deficit*: standard per-pixel binary cross-entropy optimizes grid cells independently, failing to guarantee joint multi-cell coverage over individual instances.

As illustrated in Fig. 1, faint micro targets (e.g., compact vehicles or distant surfers) easily suffer severe coarse-stage pruning in existing single-evidence selectors (e.g., ESOD [3]) due to degraded semantic contrast. To resolve these challenges, we propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a dual-evidence selective framework that couples high-frequency spectral gradients with semantic objectness and soft-coverage optimization. More specifically, our key contributions are:

1. **Dual-Evidence Spatial Selector**: We couple high-level semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ spatial complexity, reliably preserving degraded micro targets.
2. **Object-Level Soft-Coverage Loss** ($\mathcal{L}_{\text{cover}}$): We formulate a differentiable joint-coverage objective to suppress premature false-negative pruning under asymmetric selection cost.
3. **Scale-Aware Edge-Efficient Head**: We integrate an inverted residual decoupled head with scale-aware Wasserstein regression to simultaneously reduce infer-

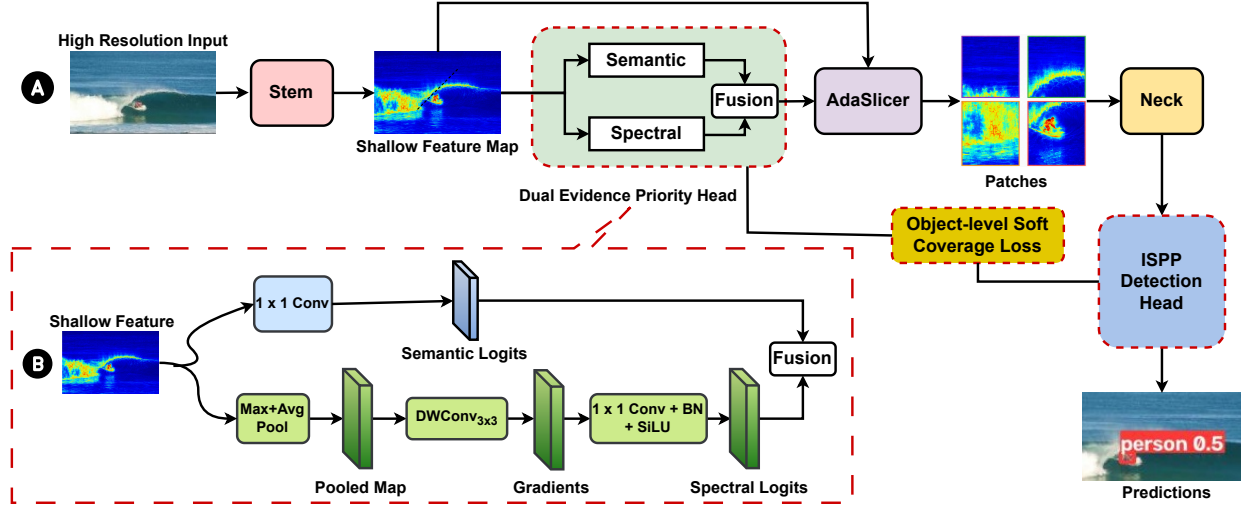


Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-evidence priority head. Red dashed boxes indicate our proposed contributions.

ence overhead and enhance micro-target localization.

2. RELATED WORK

2.1. Small Object Detection in Aerial Imagery

Detecting micro targets in high-resolution aerial and maritime vision is challenged by extreme scale disparity, low contrast, and massive background areas [1, 2, 4, 10]. Classical solutions employ density-guided patch cropping (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]) or coarse-to-fine slicing (e.g., ResBiDet [14]), which incur redundant feature extraction and complex boundary stitching. Other works explore specialized metrics, such as Normalized Gaussian Wasserstein Distance (NWD) [15], RFLA [16], and scale-aware balancing (SSAB-Net) [9]. However, dense detectors uniformly evaluate all spatial locations, leaving background redundancy unresolved.

2.2. Selective and Dynamic Computation

Dynamic computation reduces arithmetic overhead by evaluating only prospective foreground regions. AutoFocus [5] predicts focus pixels for multi-scale inference, QueryDet [6] accelerates feature pyramids via cascaded sparse queries, and CEASC [7] designs adaptive activation masks. ESOD [3] routes foreground patches (AdaSlicer) after the stem. Nonetheless, existing selectors rely strictly on single semantic activations and independent BCE loss, making them prone to early-stage pruning. In contrast, HESOD combines dual-evidence routing with joint soft-coverage optimization.

2.3. Spectral Saliency and Frequency Representations

Frequency-domain cues provide rich spatial structural priors. SET [8] demonstrated that micro objects preserve distinct high-frequency gradients even when semantic responses

fade, though it applies dense frequency operations across feature pyramids. HESOD repurposes spectral saliency as an ultra-lightweight, channel-pooled routing signal to ensure recall-safe selection at negligible overhead.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 2(a). Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow stem extracts dense feature map $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The *Dual-Evidence Spatial Selector* then proceeds in two steps: (i) a priority head fuses semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) to generate continuous priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer module dynamically crops candidate foreground patches $\mathcal{P}$ directly from $\mathbf{F}$. Downstream backbone, neck, and lightweight ISPP decoupled heads evaluate only these retained patches, supervised by an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue targets with collapsed semantic responses, HESOD couples semantic confidence with structural gradient saliency (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class logits $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_e}$. In parallel, the spectral branch extracts high-frequency structural gradients by compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where $[\cdot \parallel \cdot]$ denotes concatenation. Channel compression ensures spatial filtering complexity remains $\mathcal{O}(1)$ regardless of

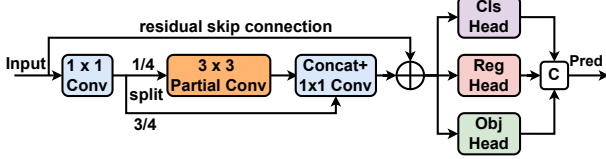


Fig. 3. Inverted Residual Decoupled Head (ISPP).

backbone width C . Learnable depthwise 3×3 convolutions (initialized via Laplacian and Sobel filters) then filter $\mathbf{F}_{\text{pooled}}$, followed by a 1×1 projection to yield spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

To prevent confidence dilution on weak targets, evidence fusion $\mathcal{F}$ satisfies *evidence preservation* $\mathcal{F}(p_s, p_f) \geq \max(p_s, p_f)$, producing priority score $s_i \in [0, 1]$:

$$s_i = \begin{cases} \max(\sigma(\mathbf{z}_i^{\text{sem}}), \sigma(\mathbf{z}_i^{\text{spec}})), & \text{Max (cluttered)} \\ \sigma(\text{Conv}_{1 \times 1}([\mathbf{z}_i^{\text{sem}} \parallel \mathbf{z}_i^{\text{spec}}])), & \text{Concat (sparse)} \end{cases} \quad (2)$$

where Max preserves salient cues in dense clutter (UAVDT), while Concat provides smooth blending in sparse scenes (SeaPerson). Candidate cells with $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance coverage: a selector may over-allocate cells to large objects while dropping tiny objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. Assuming independent cell activations, the probability that instance j is covered by at least one selected cell is:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \quad (3)$$

The soft-coverage loss directly optimizes joint instance coverage:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log(p_j^{\text{cover}} + \epsilon), \quad (4)$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ penalizes misses on micro targets ($a_j < 4$) based on box area a_j . The total training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Efficient Decoupled Head and Scale-Aware Loss

To reduce computational overhead while boosting localization precision, we incorporate an inverted residual partial-convolution (ISPP) decoupled head [17] and a Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 3, the ISPP head expands neck features to $2C_1$, partitioning them into an active branch (0.25 ratio)

processed by 3×3 Partial Convolution (PConv) and an identity branch (0.75 ratio), where $\odot$ denotes channel concatenation. After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, reducing compute while eliminating task interference.

Furthermore, standard IoU-based regression suffers from vanishing gradients on micro objects due to extreme sensitivity to tiny spatial misalignments. To ensure robust localization, SABL dynamically balances Gaussian Wasserstein distance ($\mathcal{L}_{\text{NWD}} = 1 - e^{-D_w/12}$) and Complete IoU (CIoU) via scale-dependent gating $\mu(s) = \exp(-(s/32)^6)$:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\mathcal{L}_{\text{NWD}} + (1 - \mu(s))(\ell_{\text{ctr}} + \alpha v), \quad (5)$$

where $s = \sqrt{w \cdot h}$ is the bounding box geometric scale, ℓ_{ctr} penalizes center distance, and v measures aspect consistency. For micro targets ($s \ll 32$ px), $\mu(s) \rightarrow 1$, activating Wasserstein distance to provide non-vanishing gradients under zero IoU overlap; for larger objects ($s \gg 32$ px), $\mu(s) \rightarrow 0$, smoothly reverting to standard CIoU regression.

4. EXPERIMENTS

4.1. Experiment Settings

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280×1280 px, urban vehicle surveillance across 3 classes) and **SeaPerson** [4, 10] (2048×2048 px, maritime search-and-rescue with $> 85\%$ micro targets < 32 px). Both benchmarks evaluate full-resolution images without artificial pre-cropping.

All models employ a YOLOv5m detector in PyTorch and train with SGD for 50 epochs under selective protocols [3]. Detection accuracy is assessed via COCO mAP@.5 and scale-stratified recall across bins: *Very Tiny* ($< 16^2$), *Tiny* (16^2 – 32^2), and *Small* (32^2 – 96^2). Model efficiency is measured by parameters, GFLOPs, FPS, and Bounding Patch Recall (BPR).

4.2. Main Benchmark Comparisons

Comparison with Dense Baselines. As shown in Table 1, classical dense detectors incur prohibitive arithmetic overhead on multi-megapixel vision grids due to uniform convolution. Compared to Faster R-CNN and RetinaNet on SeaPerson (2048×2048), HESOD delivers substantial accuracy gains of **+22.3 pp** and **+30.1 pp** mAP@.5, while slashing GFLOPs by **7.4×** and **4.5×** at **80.7 FPS**, confirming that selective computation is essential for edge deployment.

Comparison with Selective Baseline (ESOD). Against ESOD, HESOD achieves clear accuracy, recall, and parameter advantages across both benchmarks. On UAVDT (1280×1280), HESOD boosts mAP@.5 by **+0.7 pp** (0.392 vs. 0.385) and overall instance recovery by **+3.13 pp** (88.30%

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP @.5	mAP @.5:95	Params (M)	GFLOPs	FPS	BPR
<i>UAVDT Benchmark</i> (1280 × 1280)						
Faster R-CNN [18]	[TBD]	[TBD]	41.5	[TBD]	[TBD]	–
RetinaNet [19]	[TBD]	[TBD]	34.0	[TBD]	[TBD]	–
ESOD [3]	0.385	0.214	<u>35.9</u>	68.2	117.8	<u>0.884</u>
HESOD (Ours)	0.392	<u>0.213</u>	25.9	<u>74.9</u>	<u>102.9</u>	0.940
<i>Delta</i> (Δ)	+0.7 pp	-0.1 pp	-27.6%	+6.7	-14.9	+5.6 pp
<i>SeaPerson Benchmark</i> (2048 × 2048)						
Faster R-CNN [18]	0.551	0.246	43.3	1546.8	23.1	–
RetinaNet [19]	0.473	0.201	36.4	942.7	28.0	–
ESOD [3]	0.750	0.320	35.8	202.4	85.7	<u>0.947</u>
HESOD (Ours)	0.774	0.326	25.9	<u>209.0</u>	<u>80.7</u>	0.991
<i>Delta</i> (Δ)	+2.4 pp	+0.6 pp	-27.6%	+6.6	-5.0	+4.4 pp

All baseline results are reproduced via best reruns under identical training protocols. BPR: Bounding Patch Recall. **Bold**: best; underline: second-best.

Table 2. Module ablation on UAVDT and SeaPerson.

Configuration	mAP @.5	GFLOPs	Very Tiny	Tiny	Small	Med/ Large	Total
<i>UAVDT Benchmark</i> (1280 × 1280)							
(1) Baseline*	0.385	68.2	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	75.0	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	0.396	98.1	83.78%	87.83%	<u>97.62%</u>	64.68%	88.83%
(4) Spectral-Pooled	0.394	99.3	86.88%	92.23%	97.74%	66.43%	91.93%
(5) Dual-Concat	0.371	83.9	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	<u>0.395</u>	90.1	<u>85.69%</u>	<u>89.97%</u>	97.37%	65.99%	<u>90.36%</u>
(7) Dual-Max+SABL	<u>0.395</u>	90.1	81.74%	88.71%	96.74%	64.16%	88.69%
(8) Dual-Max+ISPP	0.394	74.9	80.76%	88.98%	97.11%	69.16%	88.83%
(9) HESOD (Full)	0.392	<u>74.9</u>	80.59%	88.13%	97.09%	<u>67.90%</u>	88.30%
<i>SeaPerson Benchmark</i> (2048 × 2048)							
(1) Baseline*	0.750	202.4	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	266.8	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	267.4	75.23%	91.69%	95.89%	86.45%	87.77%
(4) Spectral-Pooled	0.767	263.4	76.13%	91.36%	95.50%	87.10%	87.77%
(5) Dual-Concat	<u>0.772</u>	281.2	76.00%	91.50%	95.68%	80.00%	87.84%
(6) Dual-Max	0.766	255.5	75.21%	91.31%	95.69%	76.77%	87.51%
(7) Dual-Concat+SABL	0.771	263.7	<u>76.67%</u>	91.49%	94.77%	80.65%	<u>87.89%</u>
(8) Dual-Concat+ISPP	0.771	217.6	75.86%	91.25%	95.85%	87.10%	87.68%
(9) HESOD (Full)	0.774	<u>209.0</u>	77.14%	<u>91.59%</u>	95.01%	84.52%	88.11%

*Trained with standard BCE (all other arms use $\mathcal{L}_{\text{cover}}$). Scale bins: Very Tiny ($< 16^2$), Tiny (16^2 – 32^2), Small (32^2 – 96^2), Med/Large ($> 96^2$ px²). **Bold**: best; underline: second-best.

vs. 85.17%, recovering +11,700 targets) with consistent recall gains across all vehicle classes (*car*: +2.88 pp, *truck*: +4.82 pp, *bus*: +13.78 pp) and BPR reaching **0.940** (+5.6 pp) at 102.9 FPS. On SeaPerson (2048 × 2048), HESOD advances mAP@.5 by **+2.4 pp** (0.774 vs. 0.750) and BPR by **+4.4 pp** (0.991 vs. 0.947), while cutting parameters by **27.6%** (25.9M vs. 35.8M) at 80.7 FPS, establishing a superior accuracy-efficiency Pareto frontier.

4.3. Ablation Studies and Analysis

Evidence Preservation and Adaptive Fusion. Table 2 isolates module contributions and reveals that optimal routing fusion is domain-adaptive. On UAVDT, affine linear concatenation (Arm 5, 0.371 mAP) suffers from confidence dilution when one branch is faint, violating evidence preservation; switching to Elementwise Max (Arm 6) strictly

preserves high-confidence evidence, boosting mAP@.5 to **0.395** (+2.4 pp) and Total Recall to **90.36%**. Conversely, on ultra-sparse SeaPerson, Dual-Concat (Arm 5, **0.772**) outperforms Dual-Max (Arm 6, 0.766), showing that smooth feature blending excels in isolated maritime scenes. Channel-pooled spectral routing (Arm 4) preserves unpooled accuracy (Arm 3) at $\mathcal{O}(1)$ spatial complexity across both benchmarks.

Co-Design of Head and Loss via Staged Training.

Adopting the inverted residual ISPP decoupled head cuts parameters by **27.6%** (25.9M vs. 35.8M) and slashes GFLOPs (74.9 on UAVDT, 209.0 on SeaPerson). While joint end-to-end training of downstream heads can perturb shallow selector representations, staged training cleanly decouples routing from fine-grained regression, preserving peak 0.392 mAP and maintaining 102.9 FPS on UAVDT. Combining scale-aware Wasserstein regression (SABL) further enhances sub-16px localization, reaching peak *Very Tiny* recall of **77.14%** on SeaPerson.

Error Bottleneck Diagnosis. Stratified analysis confirms that target recovery is strongest on *Tiny* (+4.81 pp) and *Very Tiny* (+3.11 pp) strata. Error diagnosis reveals that dual routing and coverage optimization reduce selector-dropped false negatives from 25.6% in baseline ESOD to 19.2%, shifting the remaining bottleneck to downstream refinement.

Qualitative Comparison. As shown in Fig. 1 on UAVDT and SeaPerson, baseline ESOD misses weak-contrast micro targets due to semantic degradation, whereas HESOD’s dual-evidence routing and scale-aware regression accurately localize all tiny instances without false alarms.

5. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence selective framework for small object detection in high-resolution aerial and maritime vision. By coupling semantic objectness with channel-pooled spectral saliency at $\mathcal{O}(1)$ complexity under an evidence preservation principle and optimizing joint instance coverage via a soft-coverage loss ($\mathcal{L}_{\text{cover}}$), HESOD prevents premature false-negative pruning on micro targets. Integrated with an inverted residual partial-convolution head (ISPP) and scale-aware regression (SABL), HESOD achieves +0.7 pp mAP@.5 (+3.13 pp recall) on UAVDT and +2.4 pp mAP@.5 on SeaPerson over baseline ESOD, while slashing parameters by 27.6% with negligible FLOPs overhead.

Future work will investigate adaptive spatial threshold scheduling and extend dual-evidence selective routing to multimodal aerial perception platforms.

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